

Final Defense Deliverables

VOICE VOYAGE

Revised 5-Stage Gameplay Mechanics

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Update in this version

The original thesis overview, objectives, developmental basis, MATATAG references, and Kindergarten/Grade competency text are retained from the uploaded defense deliverable.

The old gameplay catalogue is replaced by the final 5-stage speech progression with 15 easy-to-understand gameplay mechanics.

The Kindergarten/Age 4-5 island examples are updated to show how the new mechanics can be used while preserving the existing curriculum themes and speech targets.

Thesis Overview

Voice Voyage is an AI-driven, gamified mobile application designed for early learners aged 4 to 8 to enable the automated screening and interactive practice of speech sound production. In the Philippines, early access to speech-language screening is severely restricted by a chronic shortage of specialized speech-language pathologists (SLPs), high assessment costs, and geographic barriers that concentrate services primarily in urban centers. Traditional assessments rely on manual, time-consuming transcriptions and paper-based picture cards, which can create diagnostic backlogs and induce anxiety in young children. Operating on a scalable hybrid care model, Voice Voyage addresses this gap by bridging international clinical standards with local healthcare needs—functioning as a low-cost digital assistant that automates early screening and provides structured, engaging home practice.

At the core of the system is an advanced technical architecture designed to capture and process child speech audio in real time. Using a fine-tuned Automatic Speech Recognition (ASR) engine based on the Wav2Vec2 architecture, the application converts recorded speech into phoneme-level International Phonetic Alphabet (IPA) representations. The system then calculates the Percentage of Consonants Correct (PCC) to establish overall speech severity and uses phonetic distance metrics to automatically classify production errors into four distinct clinical types: substitutions, omissions, distortions, and additions. This diagnostic engine is anchored in the ASHA Practice Portal guidelines and global developmental norms established by McLeod and Crowe (2020), ensuring that speech sound deviations are measured against empirical, chronological age-normative acquisition thresholds.

Voice Voyage translates these objective clinical evaluations into tailored learning experiences through an adaptive, gamified interface. Once a child's phonetic weaknesses are identified, a Generative AI content module generates personalized speech practice exercises dynamically mapped to the DepEd MATATAG curriculum standards, guaranteeing grade-level appropriateness.

Objectives

To develop and evaluate a mobile-based gamified speech assessment and learning application that provides personalized, curriculum-aligned speech exercises for children using adaptive AI technology and expert-guided monitoring.

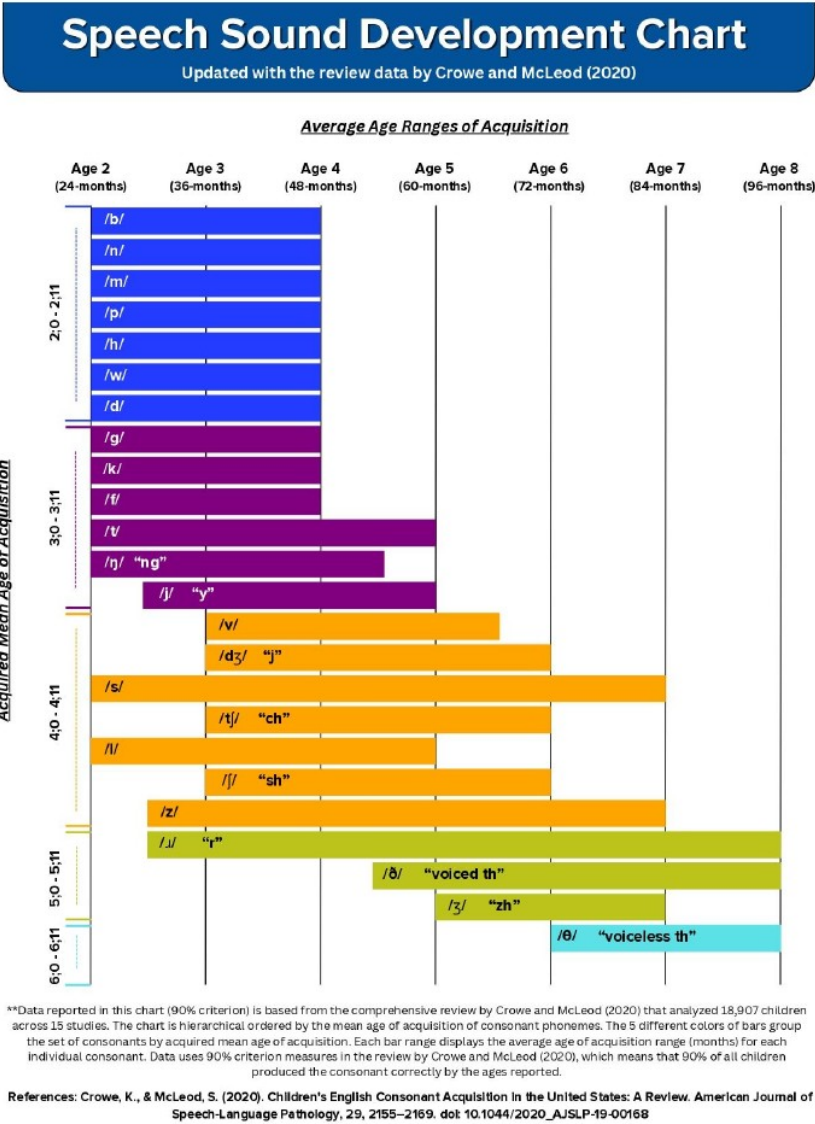
Specifically, it aims:

1. To develop an automated screening module that utilizes chronological age-normative thresholds and phonetic distance metrics to categorize speech sound errors by type.
2. To develop a module that generates personalized adaptive learning content mapped to DepEd MATATAG competencies for grade-level appropriateness based on the screening results.
3. To compare and select a model from high-accuracy Automatic Speech Recognition (ASR) algorithms for real-time phoneme-level accuracy detection.
4. To integrate the selected speech detection model into a gamified mobile interface that uses interactive mechanics to make the high-repetition practice required for phonetic habituation engaging.
5. To implement a progress-tracking feature aligned with assessment metrics for data-driven monitoring.
6. To evaluate the app's usability through feedback from speech pathologists, teachers, caretakers, and parents to ensure the platform is accessible and effective as a supportive tool.

American Speech-Language-Hearing Association (ASHA)

Guideline, Mcleod & Crowe (2018; 2020)

American Speech–Language–Hearing Association (ASHA) Guideline, Mcleod & Crowe (2018; 2020)



Dep Ed Matatag Curriculum

Matatag Curriculum Guide: Curriculum Guide

K-10 Learning Materials: Learning Materials

Grade Level	Curriculum Guide
Kindergarten	Kinder Guide
Grade 1	Language Guide / Reading and Literacy
Grade 2	Grade 2 Guide pp. 47-60
Grade 3	Grade 3 Guide pp. 61-76

The following are the essential subdomains for developing literacy in KS1:

- Oral Language
- Phonological Awareness
- Phonics
- Word Study
- Grammar Awareness
- Vocabulary
- Comprehending and Analyzing Texts
- Creating and Composing Texts

Kindergarten Learning Competencies

7. identify one's given name, friends' names, their family members, and common things they use that are found at home;

8. produce the sound of the letter it stands for;

15. use communication tools properly and technology appropriately;

23. recognize that sounding off letters form words;

Grade 1 Learning Competencies

Grade 2 Learning Competencies

Grade 3 Learning Competencies

7. identify one's given name, friends' names, their family members, and common things they use that are found at home;

8. produce the sound of the letter it stands for;

15. use communication tools properly and technology appropriately;

23. recognize that sounding off letters form words;

Grade 1 Learning Competencies

SUBDOMAINS	LEARNING COMPETENCIES	MACRO SKILLS			
		L	S	R	W
Phonological Awareness (oracy for literacy)	RL1PA-I-1 Chant nursery rhymes and poems.	L	S		
	RL1PA-I-2 Segment a two-three syllable word into its syllabic parts.	L	S		
	RL1PA-I-3 Identify rhyming words in nursery rhymes, poems, and chants.	L	S		
	RL1PA-I-4 Say two or three words that rhyme.	L	S		
	RL1PA-I-5 Identify initial sounds (vowels, consonants, and semi-vowels, if any).	L	S		
Phonics and Word Study (sounds to words)	RL1PWS-I-1 Produce the sound of the letters of L1.	L	S	R	
	RL1PWS-I-2 Identify the letters in L1.	L	S	R	
	RL1PWS-I-3 Isolate sounds (consonants and vowels) in a word (beginning and/or ending).	L	S	R	
	RL1PWS-I-4 Substitute individual sounds in simple words to make new words.	L	S	R	
	RL1PWS-I-5 Sound out words accurately.	L	S	R	

Grade 2 Learning Competencies

SUBDOMAINS	LEARNING COMPETENCIES	MACRO SKILLS			
		L	S	R	W
Phonological Awareness	EN2PA-I-1 Recognize rhymes in chants, poems, and stories heard.	L	S		
	EN2PA-I-2 Segment onset and rime.	L	S		
Phonics and Word Study (sounds to words)	EN2PWS-I-1 Identify Grade 2 level-appropriate sight words.	L	S	R	W*
	EN2PWS-I-2 Identify alphabet letter names.	L	S	R	W*
	EN2PWS-I-3 Read words accurately and automatically according to word patterns (initial, final, medial): CVC words	L	S	R	W*

Grade 3 Learning Competencies

SUBDOMAINS	LEARNING COMPETENCIES	MACRO SKILLS			
		L	S	R	W
Phonics and Word Study (sounds to words)	EN3PWS-I-1 Identify Grade 3 level-appropriate sight words.	L	S	R	W
	EN3PWS-I-2 Read words accurately and automatically according to word patterns (initial, final, medial).	L	S	R	W
	1 CVCC words	L	S	R	W
	2 CCVC words (clusters and diphthongs)	L	S	R	W
	3 VCV words	L	S	R	W
	4 VCCV words	L	S	R	W

Gameplay Mechanics

Revised final gameplay system

Voice Voyage now uses a 5-stage speech-learning progression. The 15 gameplay mechanics are organized by what the child is expected to do with the target sound, starting with listening and recognition, then moving to guided speech, independent speech, and finally contextual communication.

Stage	Main Goal	Speech Demand	Gameplay Numbers
Step 1 - Listen	Become familiar with target words and sounds.	No required production yet.	1-3
Step 2 - Recognize	Identify the correct word or target sound.	Listening/selection only.	4-6
Step 3 - Guided Say	Produce the target after a model or strong cue.	Speaking is required.	7-9
Step 4 - Independent Say	Produce the target without hearing it immediately before.	Independent speaking is required.	10-12
Step 5 - Use in Context	Use targets in phrases, sentences, roleplay, or story.	Meaningful contextual speech is required.	13-15

Core design rules

Step 1 and Step 2 prepare the child for speech. Step 3 to Step 5 require speaking for meaningful progress.

A child does not need to start at Step 1. Screening and previous performance can place the child at the most appropriate starting stage.

Words introduced in Step 1 should be reused in Step 2 and can continue into later stages so the child practices familiar targets with gradually reduced support.

Retries should be encouraging. The child keeps earned progress; the app adds a stronger hint, a slower model, or fewer distractors instead of removing lives.

Game rounds should be short, visually clear, and child-friendly. No public competition or punitive loss states.

Design inspiration: Shadowing, friendly speaking challenges, conversational practice, adaptive learning, and immediate pronunciation feedback were adapted from AI Speak: Fun English for kids. Voice Voyage uses original names, visuals, rewards, and interactions.

Step 1 - Listen

Goal: hear and become familiar with the target sound before production.

1. Picture Listen | Step 1 - Listen

Goal: Introduce 3 target words using clear pictures and model audio.

How it works: Three images appear. The child taps each image. Every tap plays the target word and gives a small visual reaction. The child must explore all three images.

Level complete: All 3 images have been tapped and heard at least once.

Speech requirement: No required speaking. This is auditory exposure only.

Example: Dog, Key, Sun. Tap DOG -> audio says 'dog'. Tap KEY -> audio says 'key'.

Implementation note: Flame can handle image reactions and scene animation. Audio playback and content configuration should come from shared services/data.

2. Sound Emphasis | Step 1 - Listen

Goal: Make the child's target sound easier to notice in the beginning, middle, or end of a word.

How it works: The app plays a model word while visually highlighting the target sound. The target sound may be stretched or repeated when appropriate for teaching emphasis, such as 'ssssun', 'kkkkkey', or emphasizing the final sound in 'dog'.

Level complete: Required target examples have been listened to.

Speech requirement: No required speaking.

Example: KEY: K glows while the model emphasizes the beginning sound. DOG: final G receives visual emphasis.

Implementation note: Use controlled pre-recorded/model prompts. Flame renders letter glow, pulse, wave, or sound-position animation.

3. Falling Sound Bubbles | Step 1 - Listen

Goal: Give repeated listening exposure through a simple moving-object game.

How it works: Image bubbles fall or float across the screen. The child taps a bubble to pop it. The pop reveals the image and plays the corresponding target word.

Level complete: All required target bubbles are popped.

Speech requirement: No required speaking.

Example: A DOG bubble falls -> child pops it -> bubble bursts -> audio says 'dog'.

Implementation note: Flame owns bubble spawning, movement, collisions/taps, particles, and pop effects. Audio/content remain shared.

Step 2 - Recognize

Goal: identify the correct word or sound after Step 1 exposure.

4. Find the Word | Step 2 - Recognize

Goal: Check whether the child recognizes a spoken target word.

How it works: The app asks a question such as 'Which one is the key?' Four pictures appear. The child selects the correct picture. Use words introduced in Step 1 whenever possible.

Level complete: 2 correct rounds are completed.

Speech requirement: No independent pronunciation required yet.

Example: Prompt: 'Which one is the dog?' Four images appear: dog, key, sun, bus.

Implementation note: Flame renders image choices and feedback. Prompt audio and target selection come from the session/controller.

5. Find the Sound | Step 2 - Recognize

Goal: Recognize a word based on the location of a target sound.

How it works: The app asks 'Find the word that starts with /k/' or 'Find the word that ends with /g/'. Four images appear. Objects may have subtle idle animation but should not reveal the correct answer.

Level complete: 2 correct rounds are completed.

Speech requirement: No independent pronunciation required yet.

Example: Prompt: 'Find the word that starts with K.' Child chooses KEY from four pictures.

Implementation note: Content must store phoneme and word-position metadata. Flame handles image choices, idle animation, and selection feedback.

6. Sound Bucket | Step 2 - Recognize

Goal: Match a heard word to the correct picture/category.

How it works: A random sound bubble plays a word. Four buckets appear below with an image and word label. The child drags the bubble into the matching bucket. When a bucket is correctly filled, it locks and cannot accept another bubble.

Level complete: 3 correct bubbles are placed.

Speech requirement: No independent pronunciation required yet.

Example: Bubble audio says 'dog'. Child drags it to the DOG bucket. The bucket fills and locks.

Implementation note: Flame handles drag/drop, bucket states, snapping, and feedback. Audio/content are supplied by the controller.

Step 3 - Guided Say

Goal: speech production begins with strong model and visual support.

7. Build & Say | Step 3 - Guided Say

Goal: Help the child construct the word first, then produce it after hearing a model.

How it works: The child completes a word using a missing sound/letter chunk. When the word is correct, the app plays the model and prompts the child to say it. Recording begins automatically or through a clear microphone cue.

Level complete: The word is correctly built and a valid speech attempt is recorded.

Speech requirement: Speaking is required, but the child hears the model immediately before the attempt.

Example: DO _ + G -> DOG -> audio: 'dog' -> prompt: 'Can you say dog?'

Implementation note: Flame handles tiles and word assembly. Flutter/services handle recording and assessment. The controller maps the result back to the game.

8. Guided Training Path | Step 3 - Guided Say

Goal: Turn repeated guided speech attempts into visible character progress.

How it works: A character follows a short course. At each obstacle, a picture and model word appear. The app says the word, then prompts 'Now you try.' A valid speech attempt allows the character to pass the obstacle.

Level complete: All configured obstacles are completed.

Speech requirement: Speaking is required with model support.

Example: KEY -> model says 'key' -> child repeats -> character jumps over a rock.

Implementation note: Flame owns character movement, obstacles, world animation, and rewards. Session controller owns prompts, recording, scoring, and progression decisions.

9. Listen, Pop & Repeat | Step 3 - Guided Say

Goal: Reuse familiar Step 1 interactions while adding required speech production.

How it works: The child taps a picture or pops a falling bubble. The target word plays. The app then says 'Your turn' and records the child's attempt.

Level complete: Required targets receive valid speech attempts.

Speech requirement: Speaking is required after the model.

Example: Pop DOG bubble -> audio: 'dog' -> 'Your turn' -> child says 'dog'.

Implementation note: Reuse Picture Listen/Falling Bubble visual components, but add the shared guided-speaking flow and microphone state.

Step 4 - Independent Say

Goal: the child produces the target without hearing the model immediately before the response.

10. Forest Discovery | Step 4 - Independent Say

Goal: Encourage independent word retrieval and production through exploration.

How it works: The child explores a forest. A glow appears under leaves, rocks, bushes, or logs. The child removes the cover and reveals an item. The app asks 'What did you find?' without playing the target word beforehand.

Level complete: 3 hidden items are found and valid independent attempts are recorded.

Speech requirement: Independent speaking is required.

Example: Leaves move aside -> KEY appears -> app asks 'What did you find?' -> child says 'key'.

Implementation note: Flame handles the environment, hiding/reveal animation, objects, and exploration. Controller handles prompt, recording, and evaluation.

11. Silly Monster | Step 4 - Independent Say

Goal: Prompt the child to independently correct a playful incorrect production.

How it works: A silly monster intentionally says the target word incorrectly. The child says the correct word. Successful productions gradually turn the monster into a friendly character.

Level complete: The configured correction rounds are completed and the friendship/progress meter is filled.

Speech requirement: Independent production is required. The correct model is not played immediately beforehand.

Example: Monster says 'tey' while showing a key -> child says 'key' -> monster becomes friendlier.

Implementation note: Keep it cooperative and non-violent. Flame handles monster states, meter, expressions, and celebration; controller evaluates speech.

12. Build the Bridge | Step 4 - Independent Say

Goal: Make independent speech directly build the game world.

How it works: A puppy, kitten, or friendly character needs to cross a broken bridge. Word bubbles appear. The child says the displayed word. A successful attempt pops the bubble and creates one bridge piece.

Level complete: All required bridge pieces are built and the character crosses safely.

Speech requirement: Independent speaking is required.

Example: Bubble shows DOG -> child says 'dog' -> bubble pops -> one bridge plank appears.

Implementation note: Flame owns word bubbles, bridge construction, character crossing, and celebration. Controller owns speech evaluation.

Step 5 - Use in Context

Goal: use the practiced target in phrases, sentences, guided conversation, and story.

13. Talk With Buddy | Step 5 - Use in Context

Goal: Move from isolated words into controlled conversational speech.

How it works: A mascot asks structured questions around the target. Prompts progress from naming an item to describing it or answering a simple follow-up. The conversation remains controlled and target-aware.

Level complete: The configured conversation turns receive valid responses.

Speech requirement: Phrase or sentence-level speaking is required.

Example: Buddy: 'What is this?' Child: 'Cat.' Buddy: 'What color is the cat?' Child: 'The cat is black.'

Implementation note: Use a controlled dialogue graph rather than unrestricted open conversation. Flutter/controller owns dialogue logic; Flame can render mascot reactions and scene animation.

14. Mission Roleplay | Step 5 - Use in Context

Goal: Use target sounds during practical communication missions.

How it works: The child receives a role-based task such as asking for an item, telling a friend what is in a bag, or reporting a discovery. The prompt is structured around approved target-rich vocabulary.

Level complete: The child completes the required communication mission.

Speech requirement: Meaningful phrase or sentence production is required.

Example: Store Mission: 'Ask the shopkeeper for a carrot.' Child: 'Can I have a carrot?'

Implementation note: Use scripted/controlled mission states. Flame renders the scenario and NPC reactions; controller validates the response and advances the mission.

15. Story Adventure | Step 5 - Use in Context

Goal: Generalize target sounds into connected, meaningful speech.

How it works: The child participates in an interactive story. Speech answers unlock the next scene. Prompts can progress from one word to a phrase, sentence, and short narration.

Level complete: The story sequence and final speaking prompt are completed.

Speech requirement: Contextual speech is required.

Example: Character finds a locked chest -> 'What does the character need?' -> child: 'A key.' Later: 'What happened?' -> child: 'The character found the key.'

Implementation note: Use a controlled branching story structure. Flame handles story scenes and character animation; controller handles prompts, recordings, assessment, and progression.

Screening-Driven Personalization

How the 15 games are selected and progressed

The key rule

The 15 games are not a fixed sequence that every child must complete from Game 1 to Game 15.

Screening determines the child's target sound/pattern and the earliest stage that still needs support.

The child can start at Step 1, Step 2, Step 3, or Step 4 depending on performance, then move forward as support is reduced.

The same target words can be reused across stages so the child experiences a clear progression from hearing to independent contextual use.

Screening/Profile Finding	Starting Stage Example	What changes
Child is unfamiliar with the target and has difficulty distinguishing it.	Step 1 - Listen	More model audio, sound emphasis, fewer words.
Child can hear the target but struggles to identify the correct word/sound.	Step 2 - Recognize	Selection and sound-position tasks.
Child recognizes the target but needs a model before saying it.	Step 3 - Guided Say	Model audio, visual cues, repeated guided attempts.
Child can imitate but struggles to say the word independently.	Step 4 - Independent Say	Remove immediate model; use pictures and game context.
Child produces the word independently and needs generalization.	Step 5 - Use in Context	Phrases, sentences, roleplay, and story.

Difficulty within each stage

Age affects presentation: younger children receive fewer choices, larger targets, shorter prompts, and stronger visual support; older children can receive longer targets, more distractors, and reduced hints.

Performance affects support: repeated difficulty increases cues or simplifies the task; stable success reduces hints and advances the child.

Difficulty does not remove the purpose of the stage. For example, Step 4 always requires independent production even if the game itself becomes visually easier.

(Kindergarten; Age 4-5) Levels and Gameplay

Updated examples using the final 15-game system

The island themes and MATATAG focus are retained from the original defense deliverable. The gameplay mechanic is now selected from the 5-stage framework. These mappings are examples; the personalization engine can change the target words and starting stage according to screening results.

Island 1: "My School & My World" (Weeks 1-2)

MATATAG Focus: Self-Identity, Name, Colors, Shapes, Spatial Positions

ASHA Target (Age 4): Early stops/nasals (/p, b, m, n/), Initial Consonant Deletion (ICD), Final Consonant Deletion (FCD), Weak Syllable Deletion (WSD)

Level	MATATAG Focus	Speech Target	New Gameplay	How it Works	Dynamic Content
1.1 Name & Letter Station	Name spelling, initial letter recognition, self-identity.	Initial consonants (/p, b, m, n/) and ICD.	Step 1 - Picture Listen	Three familiar images/letters appear. The child taps all three to hear the word and emphasized initial sound.	Default: Pen, Book, Map, Name. ICD target: Pig, Ball, Map with stronger first-sound cue.
1.2 Classroom Bag Collector	Classroom routines, sorting items by colors and shapes.	Weak Syllable Deletion and multisyllabic shapes.	Step 2 - Sound Bucket	A sound bubble plays a classroom item. The child drags it into the matching image bucket. Complete 3 correct bubbles.	Default: Book, Pencil, Paper, Crayon, Bag, Scissors. WSD target: Banana, Elephant, Paper with syllable cues.
1.3 Sound Beach Clean-Up	Spatial concepts and object classification.	FCD (/p, b, m, n, t, d, k, g/).	Step 2 - Find the Sound	The app asks for a word that ends with a target sound. Four beach-item pictures appear. Complete 2 correct rounds.	Cup, Sun, Boat, Bed, Bike, Bag. Final consonant is visually highlighted during feedback.
1.4 Syllable Bridge Crossing	Number sense 1-3, step counting, spatial navigation.	Multisyllabic integration and early stops (/t, d/).	Step 3 - Guided Training Path	The child hears a target word, repeats it, and each valid attempt lets the character cross one stepping-stone obstacle.	Desk, Ladder, Banana. Target words: Ten, Dog, Water, Ladder, Elephant.

Island 2: "My Body & My Feelings" (Weeks 3-6)

MATATAG Focus: Emotions, Body Parts, 5 Senses, Hygiene, Numbers 1-5

ASHA Target (Ages 4-5): Fronting (/k, g/ -> /t, d/), Stopping of fricatives (/s, f/ -> /t, p/), Nasal /m/ production

Level	MATATAG Focus	Speech Target	New Gameplay	How it Works	Dynamic Content
2.1 Emotion Mirror Lagoon	Identifying basic feelings (Happy, Sad, Angry, Scared, Surprised).	Stopping of fricatives (/s, f/ -> /t, p/).	Step 2 - Find the Word	The app asks for an emotion/target word and shows 4 images. Child selects the correct one; complete 2 rounds.	Default: Happy, Sad, Angry, Scared, Surprised. Target profile can use Sun, Sad, Face, Smile.
2.2 Body Parts Puzzle	Identifying body parts (Head, Hand, Foot, Leg, Knee, Ear, Eye, Nose).	Fronting (/k, g/ -> /t, d/).	Step 3 - Build & Say	Child completes the target word, hears the correct model, then repeats it into the mic.	Default body words. Fronting targets can inject Key, Goat, Coat and other approved /k,g/ words.
2.3 Senses & Hygiene Trail	The 5 senses, hygiene habits, quantities 1-5.	Fricative /f/ and nasal /m/ production.	Step 3 - Listen, Pop & Repeat	Child pops hygiene bubbles. Each word plays, then 'Your turn' prompts a spoken attempt.	Soap, Fish, Muffin, Leaf, Water, Towel. Target profile: Fish, Muffin, Leaf, Mop, Foam.
2.4 Echo Cave (Voice Studio)	Verbal self-expression and simple phrases.	Speech self-monitoring and auditory feedback integration.	Step 5 - Talk With Buddy	Buddy asks short structured questions. Child responds using simple target-rich phrases.	Default: 'I am happy', 'Wash my hands', 'See with eyes'. Target phrases can include 'See the sun' or 'Clean my coat'.

Island 3: "Family & Healthy Food" (Weeks 7-8)

MATATAG Focus: Family Roles, Small vs. Big Family, Healthy Foods (Go, Grow, Glow)

ASHA Target (Age 5): Gliding (/y, h/), Palatal Fronting (/sh/ -> /s/), Cluster Reduction (S/L-Clusters)

Level	MATATAG Focus	Speech Target	New Gameplay	How it Works	Dynamic Content
3.1 Family Tree Sorting	Family roles and family size.	Gliding target profile from the original plan.	Step 2 - Sound Bucket	A spoken family/target word appears as a sound bubble. Child drags it to the matching picture bucket. Complete 3 correct matches.	Default: Father, Mother, Sister, Brother, Baby, House. Target content can inject approved words from the learner profile.
3.2 Healthy Plate Market	Nutrition and food groups (Go, Grow, Glow).	Palatal Fronting (/sh/ -> /s/).	Step 4 - Forest Discovery (Market Theme)	Child reveals hidden market items from crates/baskets. App asks 'What did you find?' Child names the item without hearing it first.	Default: Apple, Banana, Fish, Milk, Bread, Egg. Target: Shoe, Flashlight, Fish, Shell, Shake when compatible with the activity.
3.3 Cluster Garden	Environmental objects and counting food sets 1-5.	Cluster Reduction (S-Clusters and L-Clusters).	Step 3 - Build & Say	Child assembles the blend/word first, then hears the model and says it.	Star, Spider, Plane, Blue, Spoon, Plate. Examples: S + TAR -> STAR; P + LANE -> PLANE.
3.4 Sound Path Jumper	Ordinal concepts and counting 1-5.	Consonant blend synthesis and multisyllabic integration.	Step 4 - Build the Bridge	Each independently spoken target word pops a word bubble and creates one bridge segment. Character crosses when all pieces are built.	Default: First, Second, Third, Four, Five. Blend targets: Spider, Star, Blue, Plane, Spoon.

Island 4: "Home & Story Land" (Weeks 9-10)

MATATAG Focus: Calendar Dates, Days of Week, Ordinals 1st-3rd, Story Sequencing, Numbers 0-10

ASHA Target (Ages 4-5): Multisyllabic phrase production, minimal pair discrimination, sound review

Level	MATATAG Focus	Speech Target	New Gameplay	How it Works	Dynamic Content
4.1 Calendar Castle	Days of the week, birthday dates, number counting 0-10.	Voiced/voiceless stops and /s,z/ stopping review.	Step 2 - Find the Sound	The app asks for a number/calendar word containing the target sound. Four pictures or number cards appear. Child selects the correct one.	Default: Zero-One-Ten. Target content: Sun, Seven, Six, Zebra, Bus where appropriate.
4.2 Story Sequencer (Three Little Pigs)	Story comprehension and narrative sequencing.	Multisyllabic sentence phrases and connected speech.	Step 5 - Story Adventure	Child follows the story and answers target-rich prompts to unlock the next scene.	Targets: Pig, Straw, House, Blow, Brick. Prompts can move from word -> phrase -> sentence.
4.3 Rhyme House Match	Auditory rhyming awareness and target-letter vocabulary.	Minimal-pair discrimination and contrastive sound perception.	Step 2 - Find the Word	The app plays/asks for one word and shows 4 picture choices, including the intended contrast. Child completes 2 correct rounds.	Default: Pig-Big, Sun-Bun, Bed-Red, Boat-Coat. Target contrasts: Key-Tea, Sun-Tun, Boat-Boa.
4.4 Sound Explorer Gate	Full Quarter 1 synthesis and cumulative review.	Final integration of Ages 4-5 target phonemes.	Step 5 - Mission Roleplay	The child completes short communication missions to unlock gate sections using customized target-rich phrases.	Default: 'My school bag', 'I love family', 'Count to ten'. Target passwords are customized to the child's current speech goals.

Island 5: "Mastery Island" (Post-Assessment)

Purpose: Fully interactive post-assessment embedded directly into gameplay mechanics to evaluate overall progress without formal test stress.

Level	Diagnostic Screener Objective	Speech Target	New Gameplay	How it Works	Assessment Target Items
5.1 Echo Waterfall	Re-evaluates early acquisition sounds (/p, b, m, n, t, d/), ICD, and FCD.	Early stops, nasals, ICD, FCD.	Step 4 - Forest Discovery	Hidden target items are revealed one at a time. The child names each item independently without a model immediately beforehand.	Pig, Ball, Cup, Sun, Ten, Bed.
5.2 Magic Mirror	Re-evaluates velar vs. alveolar substitution (/k, g/ vs. /t, d/).	Fronting and Backing.	Step 4 - Silly Monster	The monster intentionally says a target incorrectly. The child gives the correct word; successful corrections fill the friendship meter.	Key, Goat, Cookie, Tiger, Bike, Bag.
5.3 Fricative & Cluster Bridge	Re-evaluates Age 5 fricatives (/s, z, f, sh/) and S/L clusters.	Stopping of Fricatives and Cluster Reduction.	Step 4 - Build the Bridge	Each independent target production creates one bridge segment. The character crosses after all required words are completed.	Sun, Bus, Fish, Shoe, Spider, Star, Blue, Plane.
5.4 Grand Story Ceremony	Evaluates overall speech intelligibility and connected narrative speech fluency.	Connected speech fluency and articulation mastery.	Step 5 - Story Adventure	The child completes a short story and records final summary sentences as the last scene.	1. 'I go to school.' 2. 'I see my family.' 3. 'I play on the island.'

(Grade 1; Age 5) Levels and Gameplay

This heading is retained from the uploaded defense deliverable. The source file does not contain the Grade 1 gameplay table after this heading.